

Day 10 Student Notes Overview

Geometry Summer Credit | Right-Triangle Similarity and Weekly Review

Name:	Date:	Period:
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Today's Big Ideas

- An altitude to the hypotenuse creates smaller right triangles that are similar.
- Right-triangle similarity can create geometric mean relationships.
- Today also prepares for the weekly assessment, not a regular daily quiz.

Skills I Need to Practice

- Name the large triangle and two smaller similar right triangles.
- Use $h^2 = pq$ when an altitude splits the hypotenuse into p and q .
- Use $\text{leg}^2 = \text{hypotenuse} \times \text{projection}$ when the diagram fits.
- Review quadrilaterals, similarity, proportions, scale factor, and coordinates.

Vocabulary

- altitude, hypotenuse, projection
- geometric mean, right-triangle similarity
- weekly assessment, scale factor, midpoint, slope

Quick Reference

- Altitude formula: $h^2 = pq$.
- Leg formula: $\text{leg}^2 = c \times \text{projection}$.
- Similarity setup: matching side / matching side.
- Perimeter scales by k ; area scales by k^2 .
- Rectangle diagonals are congruent; rhombus sides are congruent.

Watch Out For

- Use right-triangle altitude formulas only when the diagram fits.
- Do not use $h = p + q$; use $h^2 = pq$.
- Do not mix up leg, projection, and full hypotenuse.
- For the weekly assessment, review older skills too.

Weekly Check Plan

The skill I feel strongest about is _____

The skill I need to review most is _____

One strategy I will use during the assessment is _____